

ANATOMY OF FLOWERING PLANTS

Complete, rigorous, exam-oriented notes covering tissue systems, internal structural variations in Dicotyledonous and Monocotyledonous roots, stems, and leaves, with key exceptions, memory tricks, and fully labeled biological diagrams.

SCOPE & SOURCE

100% NCERT Preserved

Pages 71 – 78 Complete



Table of Contents & Quick Navigation

 00. Chapter Mind Map

6.1 The Tissue System

6.1.1 Epidermal System

6.1.2 Ground Tissue System

6.1.3 Vascular System

6.2.1-2 Dicot vs Monocot Root

6.2.3-4 Dicot vs Monocot Stem

6.2.5-6 Dicot vs Monocot Leaf

 **Final Revision Summary & Key Master Matrix**





ANATOMY OF FLOWERING PLANTS (CHAPTER 6)

6.1 THE TISSUE SYSTEM

1. EPIDERMAL

Epidermal Cells
Stomata
(Guard/Subsidiary)
Trichomes & Root Hairs

2. GROUND

Parenchyma
Collenchyma
Sclerenchyma
(Cortex, Pericycle, Pith, Mesophyll)

3. VASCULAR

Xylem + Phloem
• Radial (Roots)
• Conjoint Open (Dicot Stem)
• Conjoint Closed (Monocot)

6.2 ANATOMY OF PLANT ORGANS

ROOTS

Dicot: 2-4 Xylem, Casparian Strips, Small Pith
Monocot: Polyarch (>6), Large Pith, No 2° Growth

STEMS

Dicot: Ring VBs, Starch Sheath, Sclerenchyma patches
Monocot: Scattered VBs, Cavities, No Phloem Parenchyma

LEAVES

Dorsiventral: Palisade + Spongy, Abaxial Stomata
Isobilateral: Equal Stomata, Bulliform Cells (Grasses)

6.1 SECTION 1: THE TISSUE SYSTEM

NCERT Page 71



Core Definition: Plant Anatomy & Tissue System

Anatomy is the study of the internal structure of living organisms. In higher plants, **cells** form the basic structural unit → organized into **tissues** → which form **organs** (roots, stems, leaves). Tissue systems are categorized into three major types based on their **structure and location** within the plant body.

SYSTEM 01

Epidermal Tissue System

Outermost protective envelope of the entire primary plant body.

- Epidermal Cells
- Stomata (Pores & Guard Cells)
- Epidermal Appendages (Trichomes & Root Hairs)

Location: Surface of all plant organs

SYSTEM 02

Ground / Fundamental Tissue System

Constitutes all tissues EXCLUDING epidermis and vascular bundles.

- Parenchyma (Cortex, Pericycle, Pith)
- Collenchyma (Hypodermis of stems)
- Sclerenchyma (Bundle sheath, Pericycle)
- Mesophyll in Leaves

Location: Bulk of internal body tissue

SYSTEM 03

Vascular / Conducting Tissue System

Complex tissues responsible for long-distance translocation.

- **Xylem:** Water & mineral transport
- **Phloem:** Organic food transport
- Arranged as *Vascular Bundles*

Location: Central stele & leaf veins

1. Epidermis (Primary Layer)

- Single-layered, continuous outermost layer composed of elongated, compactly arranged **parenchymatous cells**.
- Cells have a small amount of cytoplasm lining the cell wall and a **large central vacuole**.
- Outside surface is covered with a thick waxy layer called **Cuticle** which prevents water loss due to transpiration. ⚠️ **EXCEPTION: Cuticle is ABSENT in roots!**

2. Stomata & Guard Cells

- Structures present in leaf epidermis that regulate **transpiration and gaseous exchange**.
- Each stoma consists of two specialized **Guard Cells** enclosing a central **Stomatal Pore**.
- **Dicot Guard Cells**: Bean-shaped / Kidney-shaped.
- **Monocot/Grass Guard Cells**: Dumb-bell shaped.
- **Wall Thickening**: Outer walls (away from pore) are *thin*; inner walls (towards pore) are *highly thickened*.
- Guard cells contain **chloroplasts** and drive stomatal opening/closing.
- **Subsidiary Cells**: Specialized epidermal cells neighboring the guard cells.

✦ **Stomatal Apparatus = Stomatal Aperture + Guard Cells + Subsidiary Cells**

3. Epidermal Appendages

Root Hairs

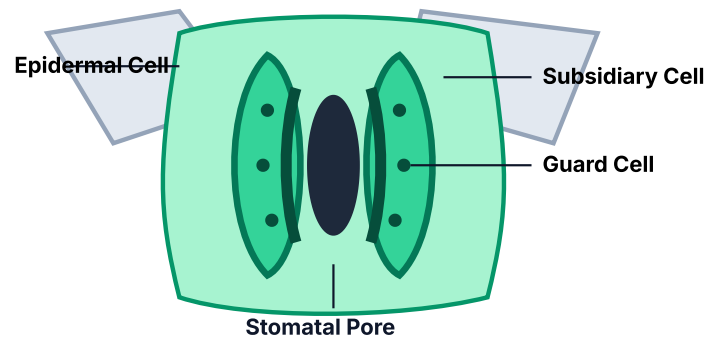
- **Unicellular** elongations of epidermal cells.
- Function: Absorbs water & minerals from soil.

Trichomes (Stem Hairs)

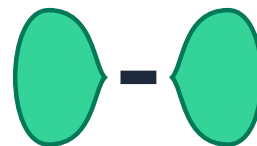
- Usually **Multicellular** in the shoot system.
- May be branched/unbranched, soft/stiff, or secretory.
- Function: Prevents water loss due to transpiration.

FIGURE 6.1 REPRODUCTION

Diagrammatic Representation of Stomata



(a) Bean-shaped Guard Cells (Dicots)



(b) Dumb-bell shaped Guard Cells (Grasses/Monocots)

6.1.2 Ground Tissue System

NCERT Page 72

Comprises all interior tissues except the epidermis and vascular bundles. Consists of **simple permanent tissues**:

Parenchyma: Present in Cortex, Pericycle, Pith, and Medullary Rays in primary stems and roots.

Collenchyma & Sclerenchyma: Provide mechanical support in hypodermis and pericycle.

Mesophyll in Leaves: Thin-walled, chloroplast-containing parenchymatous ground tissue responsible for photosynthesis.

6.1.3 Vascular Tissue Types

NCERT Page 73

1. Radial Arrangement (Roots):

Xylem and Phloem are arranged in an *alternate manner along different radii*.

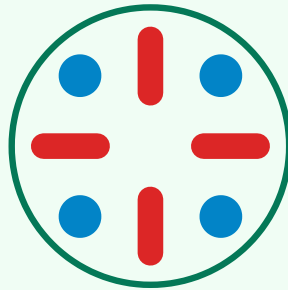
2. Conjoint Open (Dicot Stems):

Xylem and Phloem situated along the *same radius*. **Cambium present** between them → possesses ability to form secondary xylem/phloem.

3. Conjoint Closed (Monocots):

Xylem and Phloem along same radius, but **No Cambium present** → Cannot undergo secondary growth.

Figure 6.2 — Types of Vascular Bundles (Radial, Conjoint Closed, Conjoint Open)



(a) Radial (Roots)

Xylem (Red) & Phloem (Blue) on alternate radii



(b) Conjoint Closed (Monocot Stem)

Phloem outer, Xylem inner, NO Cambium



(c) Conjoint Open (Dicot Stem)

Phloem + Cambium Strip (Yellow) + Xylem

6.2 SECTION 2: ANATOMY OF DICOTYLEDONOUS & MONOCOTYLEDONOUS ROOTS

NCERT Pages 73-74

Anatomical Comparison Matrix: Dicot Root vs. Monocot Root

Anatomical Feature	Dicotyledonous Root (e.g., Sunflower)	Monocotyledonous Root
1. Epiblema (Piliferous Layer)	Outermost single layer; bears unicellular root hairs . Cuticle absent.	Same as dicot root (outermost with unicellular root hairs).
2. Cortex	Multiple layers of thin-walled parenchyma with intercellular spaces.	Similar to dicot root (multi-layered parenchymatous cortex).
3. Endodermis & Suberin	Single layer barrel-shaped cells. Tangential & radial walls have Casparian strips (waxy suberin deposition).	Same as dicot root (Endodermis present with Casparian strips).
4. Pericycle	Thick-walled parenchymatous cells. Gives rise to lateral roots & vascular cambium during 2° growth.	Gives rise to lateral roots ONLY. No cambium formed!
5. Vascular Bundles & Number	Radial arrangement. Diarch to Tetrarch (Usually 2 to 4 Xylem & Phloem patches).	Radial arrangement. Polyarch (Usually more than 6 xylem bundles).
6. Pith	Small or inconspicuous.	Large and well-developed.
7. Conjunctive Tissue	Parenchymatous cells between xylem & phloem; forms cambium ring later.	Parenchymatous cells lying between xylem and phloem.
8. Secondary Growth	PRESENT . Undergoes secondary growth via vascular & cork cambium.	ABSENT . Monocot roots never undergo secondary growth.

🧠 **Memory Trick for Root Layers (Outside → Inside):**

"Every Cat Eats Pork Cutlets Pretty Very Patiently"

Epiblema → Cortex → Endodermis (Casparian) → Pericycle → Conjunctive Tissue → Vascular Bundles → Pith

★ **High-Yield Exam Point: Stele Definition**

All plant tissues situated on the **INNER SIDE of the endodermis** collectively constitute the **STELE**.

✚ **Stele = Pericycle + Vascular Bundles + Pith + Conjunctive Tissue**

6.2 SECTION 3: ANATOMY OF DICOTYLEDONOUS & MONOCOTYLEDONOUS STEMS

NCERT Pages 74-76

Dicotyledonous Stem Anatomy

Ring Arrangement

- 1. Epidermis:** Outermost protective layer covered with a thin cuticle. Bears **trichomes (multicellular stem hairs)** and a few stomata.
- 2. Cortex (3 Sub-zones):**
 - *Outer Hypodermis:* Few layers of **collenchymatous cells** below epidermis; provides mechanical strength.
 - *Cortical Layers:* Rounded thin-walled parenchymatous cells with intercellular spaces.
 - *Endodermis (Starch Sheath):* Innermost layer, rich in starch grains.
- 3. Pericycle:** Present on inner side of endodermis above phloem as **semilunar patches of sclerenchyma**.
- 4. Medullary Rays:** Radially placed parenchymatous cells between vascular bundles.
- 5. Vascular Bundles:**
 - Arranged in a **characteristic RING**.
 - **Conjoint, Open** (cambium present), with **Endarch Protoxylem**.
- 6. Pith:** Large, central parenchymatous zone with prominent intercellular spaces.

Monocotyledonous Stem Anatomy

Scattered Arrangement

- 1. Epidermis:** Outermost layer covered with cuticle. Epidermal hairs/trichomes usually absent.
- 2. Hypodermis:** Composed of **sclerenchymatous cells** (unlike dicot stem's collenchyma).
- 3. Ground Tissue:** Large, conspicuous, undifferentiated **parenchymatous mass**. No distinct cortex, endodermis, or pith!

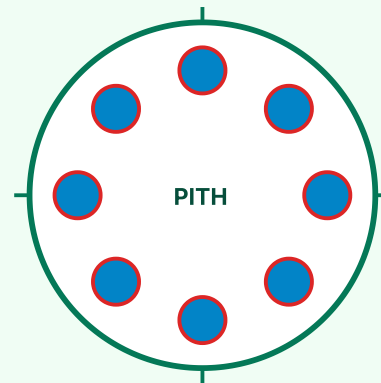
4. Vascular Bundles:

- **Scattered** throughout the ground tissue.
- Each bundle surrounded by a **sclerenchymatous bundle sheath**.
- **Conjoint and Closed** (no cambium, no secondary growth).
- *Peripheral bundles* are smaller than *centrally located bundles*.

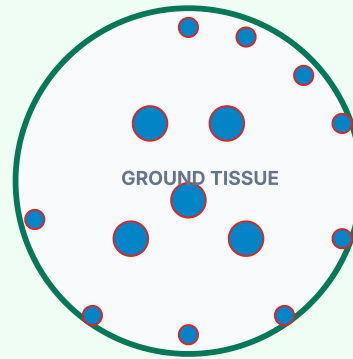
⚠ CRITICAL EXAM EXCEPTIONS

1. **Phloem Parenchyma is ABSENT** in monocot stems!
2. **Water-containing cavities** are present within the vascular bundles (formed by disintegrating protoxylem).

Figure 6.4 Representation — Dicot Stem Ring vs Monocot Stem Scattered Layout



(a) Dicot Stem: Vascular Bundles in a RING



(b) Monocot Stem: SCATTERED Vascular Bundles

6.2 SECTION 4: ANATOMY OF DORSIVENTRAL (DICOT) & ISOBILATERAL (MONOCOT) LEAVES

NCERT Pages
76-77

6.2.5

Dorsiventral (Dicotyledonous) Leaf

1. Epidermis:

- Covers both upper surface (**Adaxial Epidermis**) and lower surface (**Abaxial Epidermis**).
- Conspicuous cuticle present.
- *Abaxial epidermis* bears **MORE stomata** than adaxial epidermis (Adaxial may even completely lack stomata!).

2. Differentiated Mesophyll (Parenchyma):

- **Palisade Parenchyma:** Adaxially placed, elongated vertically oriented parallel cells rich in chloroplasts.
- **Spongy Parenchyma:** Oval or round, loosely arranged below palisade extending to lower epidermis. Contains numerous large air spaces & cavities.

3. Vascular System:

- Bundles seen in veins and midrib. Size depends on vein size (varies due to **reticulate venation**).
- Each bundle surrounded by a layer of thick-walled **Bundle Sheath Cells**.

6.2.6

Isobilateral (Monocotyledonous) Leaf

1. Equal Stomatal Distribution:

- Stomata are present on **BOTH surfaces** of the epidermis (Amphistomatic).

2. Undifferentiated Mesophyll:

- Mesophyll is **NOT differentiated** into palisade and spongy parenchyma. Cells are uniform.

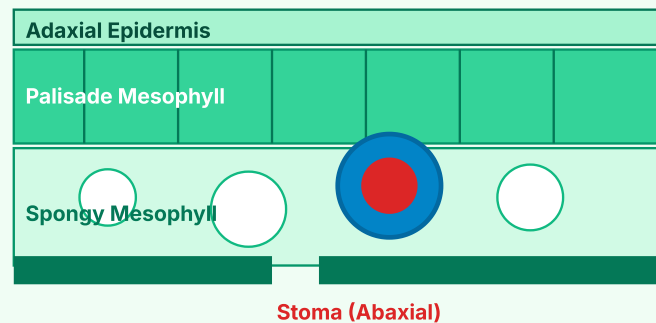
3. Bulliform Cells (Specialized Grass Adaptation):

- Certain adaxial epidermal cells along veins modify into *large, empty, colourless cells* called **Bulliform Cells**.
- **Turgid State (Water Absorbed)**: Leaf surface is exposed/flat.
- **Flaccid State (Water Stress)**: Causes leaf to **curl inwards** to minimize water loss!

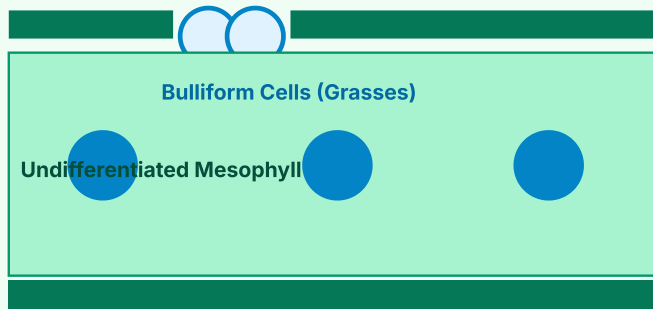
4. Uniform Vascular Bundles:

- Near similar sizes of vascular bundles due to **parallel venation** (except in main veins).

Figure 6.5 — Anatomical Cross Sections of Leaves



(a) Dorsiventral Leaf (Dicot) — Differentiated Mesophyll



(b) Isobilateral Leaf (Monocot) — Bulliform Cells & Equal Stomata



FINAL REVISION — MASTER COMPARATIVE MATRIX

Quick-recall cheat sheet summarizing all 6 anatomical organs

Organ	Outermost Layer	Hypodermis	Ground Tissue / Cortex	Vascular Bundles	Pith	Secondary Growth
Dicot Root	Epiblema (Unicellular hairs)	Absent	Parenchymatous Cortex + Endodermis (Casparian Strips)	Radial, Diarch to Tetrarch (2-4 bundles)	Small / Inconspicuous	Present
Monocot Root	Epiblema (Unicellular hairs)	Absent	Parenchymatous Cortex + Endodermis (Casparian Strips)	Radial, Polyarch (>6 bundles)	Large & Well-developed	Absent
Dicot Stem	Epidermis (Cuticle + Trichomes)	Collenchymatous	Cortex + Starch Sheath Endodermis + Sclerenchyma Pericycle	Conjoint, Open, Ring Arrangement, Endarch	Large & Parenchymatous	Present
Monocot Stem	Epidermis (Cuticle)	Sclerenchymatous	Undifferentiated Parenchyma (No distinct cortex/pith)	Conjoint, Closed, Scattered, Bundle Sheath present	Absent (Merged in ground tissue)	Absent
Dorsiventral Leaf (Dicot)	Epidermis (Abaxial stomata > Adaxial)	N/A	Differentiated into Palisade & Spongy Mesophyll	Conjoint, Closed, Bundle sheath surrounded	N/A	Absent
Isobilateral Leaf (Monocot)	Epidermis (Stomata equal on both sides + Bulliform cells)	N/A	Undifferentiated Mesophyll	Conjoint, Closed, Near similar sizes	N/A	Absent



Complete NCERT Textbook Exercise Solutions

Official, fully answered questions from NCERT Page 78

Q1. Draw illustrations to bring out the anatomical difference between: (a) Monocot root and Dicot root (b) Monocot stem and Dicot stem.

Answer Summary & Key Differences:

(a) Root Differences: Dicot roots possess diarch to tetrarch (2-4) xylem patches with a small/inconspicuous pith and undergo secondary growth. Monocot roots possess polyarch (>6) xylem patches with a large, well-developed pith and do NOT undergo secondary growth.

(b) Stem Differences: Dicot stems have collenchymatous hypodermis, ring-arranged open vascular bundles, sclerenchymatous pericycle patches, and a large pith. Monocot stems have sclerenchymatous hypodermis, scattered closed vascular bundles with bundle sheaths, water cavities, and no phloem parenchyma.

Q2. Cut a transverse section of young stem of a plant from your school garden and observe it under the microscope. How would you ascertain whether it is a monocot stem or a dicot stem? Give reasons.

Answer Key Identification Criteria:

- **If Dicot Stem:** Vascular bundles are arranged in a distinct **RING**; bundles are *open* (cambium present); hypodermis is *collenchymatous*; distinct central *pith* present.
- **If Monocot Stem:** Vascular bundles are **SCATTERED** throughout ground tissue; bundles are *closed* with *bundle sheaths*; hypodermis is *sclerenchymatous*; water cavity present in bundle.

Q3. The transverse section of a plant material shows the following anatomical features — (a) the vascular bundles are conjoint, scattered and surrounded by a sclerenchymatous bundle sheath. (b) phloem parenchyma is absent. What will

you identify it as?

Identification Result: Monocotyledonous Stem

Reasoning: Scattered vascular bundles surrounded by sclerenchymatous bundle sheaths and the complete absence of phloem parenchyma are diagnostic hallmarks exclusively found in monocot stems.

Q4. What is stomatal apparatus? Explain the structure of stomata with a labelled diagram.

Definition: The **stomatal apparatus** consists of the *stomatal aperture (pore)*, two surrounding *guard cells*, and adjacent specialized *subsidiary cells*.

Structure: Stomata are microscopic pores in the leaf epidermis regulating gaseous exchange and transpiration. Each pore is flanked by two guard cells (bean-shaped in dicots, dumbbell-shaped in grasses). Outer walls of guard cells are thin while inner walls are thick. Guard cells contain chloroplasts that regulate pore opening/closing via turgor changes.

Q5. Name the three basic tissue systems in the flowering plants. Give the tissue names under each system.

1. **Epidermal Tissue System:** Epidermal cells, Stomata (guard cells, subsidiary cells), Epidermal appendages (root hairs, trichomes).
2. **Ground / Fundamental Tissue System:** Parenchyma, Collenchyma, Sclerenchyma (Cortex, Pericycle, Pith, Medullary rays, Mesophyll).
3. **Vascular Tissue System:** Complex tissues — Xylem (vessels, tracheids, xylem fibres, xylem parenchyma) and Phloem (sieve tubes, companion cells, phloem fibres, phloem parenchyma).

Q6. How is the study of plant anatomy useful to us?

Significance of Plant Anatomy:

1. Helps understand structural adaptations of plants to diverse ecological environments.
2. Facilitates distinction between monocotyledons and dicotyledons.
3. Essential for commercial wood identification, timber quality evaluation, and detecting adulteration in medicinal plant products.

4. Provides insights into evolutionary relationships among plant taxa.

Q7. Describe the internal structure of a dorsiventral leaf with the help of labelled diagrams.

Main Structural Components:

1. **Epidermis:** Covers upper (adaxial) and lower (abaxial) surfaces. Abaxial bears more stomata. Cuticle present.
2. **Mesophyll:** Differentiated into *Palisade parenchyma* (upper, elongated vertically parallel cells rich in chloroplasts) and *Spongy parenchyma* (lower, round, loosely arranged with air cavities).
3. **Vascular System:** Vascular bundles located in midrib and veins, surrounded by thick-walled bundle sheath cells. Xylem towards adaxial side, Phloem towards abaxial side.